Lecture 4

Tuesday, 24 January 2023 12:01 PM

Intrinsic Kinetics in Catalysis

Active centers ; Homogenous catalysis → Active centers in the bulk

Heterogenous catalysis → " on surface

$$A + B \overset{Cat.}{\rightleftharpoons} C + D$$

$$\left.\begin{aligned} A + X_1 &\rightleftharpoons AX_1 \\ B + X_2 &\rightleftharpoons BX_2 \end{aligned}\right\} \text{Chemisorption}$$

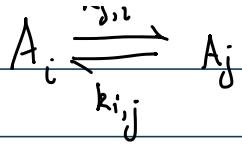
$$AX_1 + BX_2 \rightleftharpoons C + D + X_1 + X_2 \quad \left.\right\} \text{Surface reaction \& desorption}$$

$$\left.\begin{aligned} P + f X_{1,2} &\rightleftharpoons 0 \\ f' X_{1,2} &\rightarrow \bar{X}_{1,2} \end{aligned}\right\} \text{Poisoning (Thermal deactivation)}$$

Lumping in Kinetic Systems

Petroleum → lumped into 4 major components: Paraffins, Olefins, Naphthalene, Aromatics } PONA (Also called "Naphtha")

$$A_i \underset{k_{i,j}}{\overset{k_{j,i}}{\rightleftharpoons}} A_j$$



$\sum_{i=1,\, i \neq j}$; A_i going to all different species (consumed will have →)

Reducing no. of species to deal w/ ⟶ Lumping